

The Introvert and the Extrovert Aspects of the Marathi House

Narendra Dengle

As an architect when I observe a house, I am drawn into examining its three basic aspects. First and foremost is the quality and character of its space. Then I try and relate this spatial character to the character of its owner—the family. These two observations then prompt me to study the culture of the society that the family belongs to.

The first area of observation, namely the spatial character of a house, is of a pure and abstract nature. It has to do with the geometry of the house, the scale and size of the spaces that it holds, its tactile quality, the quality of light within, its relationship with the natural as well as man-made surroundings, its visual and plastic form, the construction technique used and the materials employed for construction.

It is imperative that these pure and abstract criteria then be examined against the socio-cultural character of the family the house is meant for. The cultural background of the family, its traditional occupation and source of income, as well as the interrelationships of the family members and their relationship with the neighbourhood and society at large, are some of the issues which are studied in this context.

The study of the two above mentioned fields then leads one to examine the whole issue of culture. Under the umbrella of culture one studies the immeasurable and the measurable aspects of human activity in a society. 'Culture is something that has to do with the religious and economic goals and social and moral attitudes. It is also an unwritten code that defines, establishes and forges human relationship.'¹ One therefore examines the process adopted by a society through centuries

to refine, correct, amend, discard and modify its goals and priorities in life. The legends, myths, rituals and way of day-to-day living which take shape due to one's response to the forces of nature, climatic changes, the characteristics of the soil and water, and one's own inner spiritual beliefs—all come under observation when the field of culture is examined. In the context of a residential dwelling—the study of how the house responds to functional and climatic conditions on the one hand and spiritual and religious beliefs on the other provides substance and philosophical content upon which the pure and abstract visual form is built.

The state of Maharashtra can be conveniently divided into five basic geographical regions—Konkan, Desh, Khandesh, Marathwada and Vidarbha. Bombay, being a major metropolis, needs a different treatment. I intend to present two case studies confined to Desh and Konkan, through which I hope to highlight the issues that one studies as an architect. I shall also make an attempt to see if through these two case studies one gets a closer view of the introvert and extrovert aspects of the Marathi house.

Marathi consciousness can be felt from the pre-Dnyaneshvar period. Dnyaneshvar's work—*Bhavarthadipika* (A.D.1290) is a rare example of the use of a mature and melodious language, i.e. old Marathi. References to the conception of the Marathi house can be found in his works, too.² Hence, although as a part of a vast region of the Deccan that included Karnatak and Telangana—Andhra, the Marathi house must have existed and begun to take shape in the late thirteenth century. But what we refer to as the Marathi house here, assumed a tangible form with the rise of the Peshwas to power. It was during their stay at the helm of the Maratha kingdom from the late seventeenth century to 1818, when the city of Pune was surrendered to the British, that the *wada* architecture came to full prominence. It was an architectural expression that could be suitably interpreted for the poor and the rich alike based on their own demands, without losing its core characteristics.

The Desh House

The *wada* as a residential archetype became an acceptable solution for various reasons. Firstly it was derived from norms recommended by the ancient Hindu treatise on architecture, the *Vastushastra*. It was a wholly introvert arrangement of rooms and verandahs around courtyards. The provision of courtyards open to the sky within ensured plenty of light and

ventilation without sacrificing privacy, while the thick walls with few and small openings on the outside provided strong security against intruders. In a neighbourhood, this type of module with rooms and verandahs around a courtyard could be closely packed with the adjoining *wadas*—a system which proved to be the most efficient and economic way of utilizing land for housing. Figures 1-8 show the different permutations of the three basic components of the *wada*: the built-up area (rooms etc.), the verandah (which was a transitional space), and the courtyard open to the sky. In the case of the larger *wadas* for wealthy and important people, the courtyard module was multiplied twice, thrice, or four times to address other larger issues such as servants' quarters and the stable for horses, and giving more privacy to the women of the household. While the outer courts were meant for receiving visitors and hence were formal, the rear courts were used as stables or servants' quarters and the middle courts were available for private domestic use. This created a series of layers of insulation between the individual and society. With the essential components of the house being the rooms, the verandah, and the courtyard, their simple and geometric permutations gave rise to an architecture of simplicity of form and richness of space. The space was rich because of the quality of light that it enjoyed, the visual layers of spaces created by the courtyards and the multipurpose uses that the courtyard was put to. The courtyard was a space that could be interpreted by every user in his own way and could also be used collectively for specific activities and rituals. Women used it for their gatherings and functions in the days when women rarely socialized outside the house. It was also used for drying grain and spices, work which was done entirely at home.

Some of the essential features of a *wada* house can be seen in Figures 9 & 10. The architectural elements of a central courtyard were the plinth (*ota*), verandah (*sopa*), terrace (*gachchi*), staircase, steps (*jina*), built tank or pond (*pushkarni* or *haud*), well (*vihir*), fountain (*karanje*), basil plant shrine (*tulsivrindavan*) and a garden of plants and flowering shrubs. The scale of the *wada* was dictated by the permutations and choice of these elements.

The flexibility and patterns in design came with the details of construction, textures of surfaces and the involvement of specialist craftsmen from different parts of the country. The craftsmen usually adorned the doorways, the *meghadambari*, or the fanciful balcony that resembled a caparison on an elephant. The brackets supporting the balcony were also beautifully carved. Decorative arts, which really were

the privilege of the rich, found expression in stone columns, capitals and bases, multifoil arches in stone and timber, in marble, granite and *shaligram* idols, and in the woodwork for posts and beams, joists and ceiling. This decoration was very simple and humble in comparison with the decoration in the elaborately decorated *havelis* of Gujarat and Rajasthan or the grandeur and expansiveness of the Mughal mansions in the north or the Deccan. It also so happened that the craftsmen who came to be employed in such work came mostly from Rajasthan.

The Maharashtrian house has been, architecturally speaking, austere in style, firm in its core architectural elements such as the particular arrangements of courtyards and modest in scale and size. The only exceptions are the *wadas* built by the Peshwas and their sardars, treasurers or bankers, as these were magnificent in scale and size. The palace-fort Shaniwarwada built by the Peshwas at Pune was a *satkhandi* or seven-storeyed structure. Usually the common man's house was either *dukhandi*, *teenkhandi*, or *chowkhandi*. i.e. two, three, or four-storeyed respectively.

The single courtyard *wadas* had little privacy for the womenfolk in comparison with the double or triple courtyard *wadas*. Since everything, like consultation, meetings or negotiations with outsiders, took place in one of the rooms facing the courtyard and all other activities remained unchanged, naturally the gap between the public and private spaces was narrowed. For instance, where meetings with outsiders took place in the drawing room, right above the entrance i.e. the *diwankhana*, the rest of the house would still be secluded from the public eye. The double courtyard house created a layer of insulation, functionally speaking, and a layer of visual elements, spatially speaking. A house therefore did not remain a block. On the contrary, it gave a sense of depth, not dissimilar to the inward reality of a human being, which is different from his physical appearance. The outside space, comprised of the courtyard surrounded by a colonnade of timber posts and beams and rooms behind, became the public space. The common man's *wada* did not even have a facade as understood in Western architecture. It had a very narrow, flat, unpretentious, functional, and rather unselfconscious disposition. The more important people had decorative balconies overlooking the streets. These were more conscious efforts at the participation of the house in a streetscape. The house in Desh therefore appears to be much more introvert in character, as if constantly engrossed in introspection, soul-searching, and content in the process. It was also an indication that the family was rather a closely knit unit. Because the office of the patriarch

was located around the front courtyard, everyone and most of the activities came under his watchful eye. The double courtyard houses indicate that women had their own independent world, which was adequately insulated from the men folk, if required. The rear entry ensured its total independence from the front section of the house where business matters with outsiders were dealt with. In a double courtyard house, space was available to play, to hold rituals and to carry on kitchen-related activities. Some of the *wadas* have temples within their premises. Smaller *wadas* had small niches or kiosks which worked as shrines. Since there was no need to have any collective worshipping in the Hindu religion, smaller shrines at the domestic level were found to be perfectly suited to the individualistic way of life.

The houses on the Desh side were composite structures having wooden frames and thick mud, brick or stone walls. The walls at the entrance and *diwankhana* were sometimes painted with mythological themes and the ceiling of the *diwankhana* had decorative woodwork. The roofing was usually in fired clay pot tiles and was laid in a slope. The flooring at the first floor level and above was in mud over wooden board or stone slabs, laid over wooden joists supported over beams and finished by being plastered with mud, cow dung or stone slabs. The courtyard flooring and the surrounding plinth were usually made of stone. The verandah had stone slabs or mud flooring depending on what the owner could afford. The glare of light was considerably reduced before it could enter the house, as the light bounced off these courtyards surrounded by two-storeyed walls. The windows were narrow but tall. The double leaf shutters were in two parts. The lower part opened from the floor level, where one sat with folded legs, while the upper part could be opened at one's eye level when one was in a standing position. The swing was the recreational element and feature of the verandah. It provided a peculiar rhythmic sound, apart from its movement, and could be heard throughout the house. The *tulsivrindavan*, the well or *pushkarni* and the *jhopala* or swing—all were not only visual elements of the ensemble but also provided the audio notes which forged strong and long-term associations with the house and remained in the memory forever. At the *tulsivrindavan*, usually at the time of god worshipping (*puja*), some recitation took place. The *pushkarni* introduced the sound of water: if it had a fountain then the sound of lightly sprinkling water, or, in case of a well, the sound of the wheel lowering the bucket into the water, then the splash, and then the bucket being pulled up. The swing was most intimately related to the desire to sing songs. Metaphorically

as well as actually, it gave a momentum to the urge to sing, to be friends with others or to give way to one's own creative impulse. All these sounds must be considered an integral part of the house. A house is after all not a mere series of walls and openings, as it is being thought of more recently in the conception of blocks of flats, but it was a place of belonging which had a great deal to contribute to one's psyche, personality, and perception of life as one grew. The visual and audio experiences therefore had a great contribution in building these faculties. A *wada* could not be imagined without the courtyard, the *pushkarni*, the rhythmic note of the swing, and the colonnade of the timber posts. These, as it were, symbolized the introverted aspect of the Marathi house of the Desh region, with a spatial rather than formal outlook.

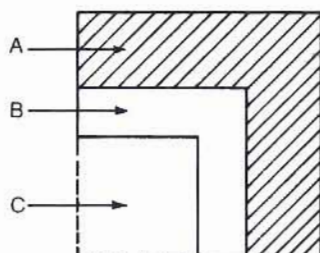
The form of the house was true to its structural system and its three-dimensionality was a direct product of the plan itself. In this respect, it appears to hold the values which would be held high in Modern Architecture. Modern Architecture, which did away with Baroque decoration and was loyal to the Classical Orders while embracing functionalism, structuralism, and, of course, industrialism, stood firm for a no-nonsense attitude in arriving at the aesthetics of architecture. Modern Architecture advocated a form which would be true to its function, true to the qualities of the material used in construction (i.e. not disguised under some fanciful decorative layers of plaster and mouldings) and a three-dimensional expression which was embedded in the plan itself. Despite the fact that Modern Architecture is a term applied to the architecture of the post-Industrial Revolution era and therefore addresses itself to many complex and urban phenomena, it can be said that the *wada* architecture of Maharashtra with its simplicity, structuralism and true-to-material approach devoid of over-decorativeness does seem to hold the seeds of the values of Modern Architecture.

A. The Typology of Houses according to *Matsyapurana*

The typology of *wada* architecture appears to have roots in the Vedic period. The Puranas, it is believed, are our main source of information for the history of the earliest period, and for the period before the sixth century B.C. Definite and detailed references and guidelines are found in the Puranas on different types of houses which could be built for different castes and men of social and political positions within the society. Guidelines on matters ranging from how to select the site and examine the soil for building the house to how to select construction

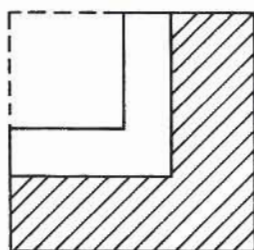
PERMUTATIONS OF THREE BASIC COMPONENTS

(A = ROOMS, B = VERANDAH, C = COURTYARD)



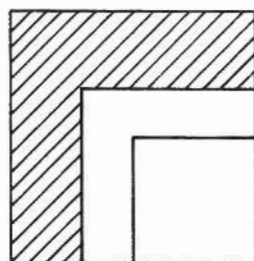
A HOUSE WITH TWO WINGS
IS CALLED 'DWISHAL'/'DUSOPI'
(द्विशाल/दुसोपी)

FIG. 1



A HOUSE HAVING ROOMS ON EAST
AND SOUTH IS KNOWN
AS 'DHANAKHYA'
(धनाख्य)

FIG. 2



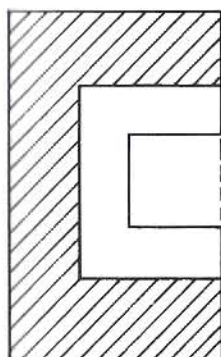
A HOUSE HAVING ROOMS
ON NORTH AND WEST IS
KNOWN AS 'YAMASURYA'
(यमसूर्य)

NORTH



FIG. 3

HOUSES WITH INNER COURTYARDS ENCLOSED ON THREE SIDES BY VERANDAHS (TRISHAL: त्रिशाल)



A HOUSE HAVING NO WING TO ITS EAST IS KNOWN AS 'SUKSHETRA' (सुकशेत्र)

FIG. 4

A HOUSE WITHOUT ITS WESTERN WING IS KNOWN AS 'PAKSHAGHNA' (पक्षघ्न)

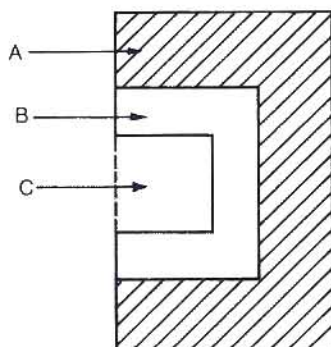
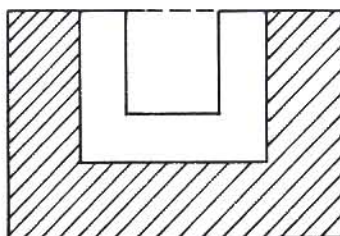


FIG. 5



A HOUSE WITHOUT ITS NORTHERN WING IS KNOWN AS 'DHANYAKA' (धान्यक)

NORTH



FIG. 6

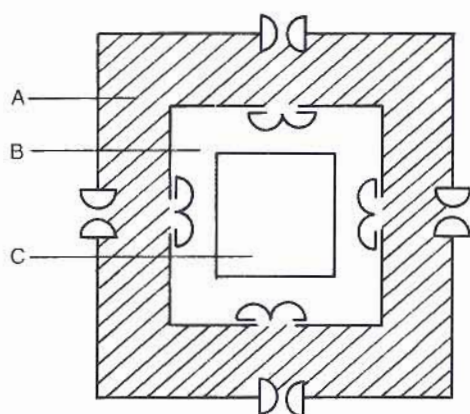
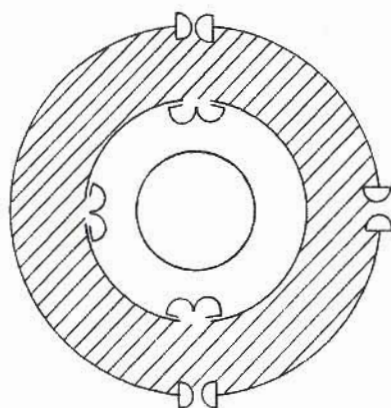


FIG. 7

- A COURTYARD HOUSE WITH FOUR WINGS FACING EACH OTHER & WITH FOUR DOORS IS CALLED 'CHATUSHALA/CHAUSOPI/BHADRA'
(चतुःशाल/चौसोपी/भद्र)
- IF IT IS WITHOUT AN EAST DOOR IT IS CALLED 'SWASTIK'
(स्वास्तिक)
- IF IT IS WITHOUT A NORTH DOOR IT IS CALLED 'RUCHAK'
(रुचक)
- IF IT IS WITHOUT A SOUTH DOOR IT IS CALLED 'VARDHAMAN'
(वर्धमान)



A CIRCULAR PALACE/TEMPLE WITHOUT A WEST DOOR WITH A VERANDAH IS CALLED 'NANDYAVARTA'
(नन्दावर्त)

NORTH



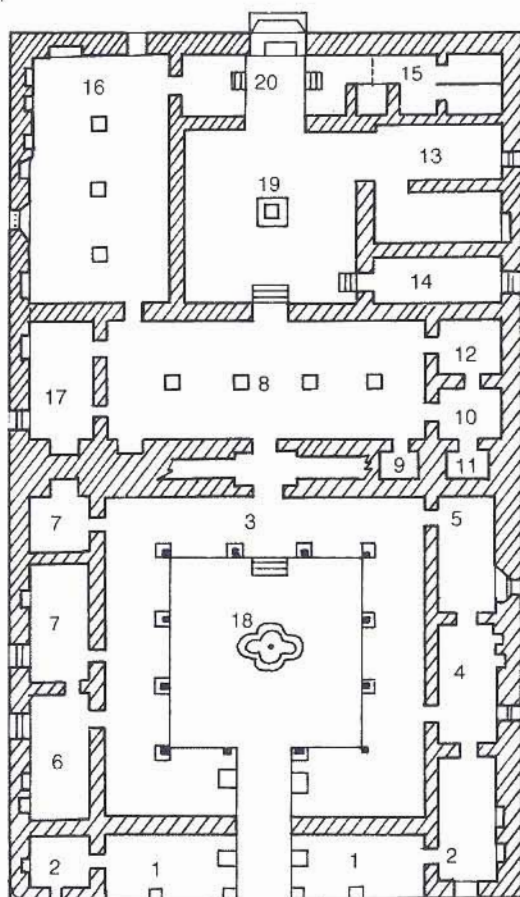
FIG. 8

SOURCE: MATSYAPURANA

FIG. 1-8 (DIAGRAMS BY AUTHOR FROM DESCRIPTIONS IN MATSYAPURANA)

PLANS OF TYPICAL WADA ARCHITECTURE SHOWING ITS INTROVERT QUALITY

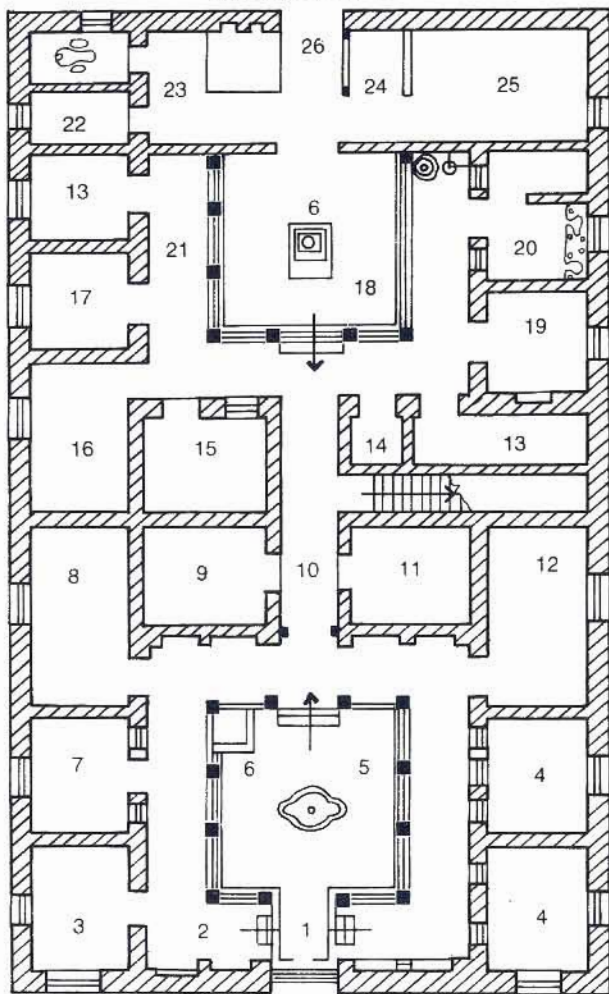
TYPICAL WADA - 1



- | | | |
|------------------------|-------------------------|------------------------------|
| 1. OSARI (VERANDAH) | 8. MANJGHAR (MIDDLE RM) | (DELIVERY RM) |
| 2. DEWADI (GUARD RM) | 9. BALAD (GRAIN STORE) | 15. NAHANI/SANDAS |
| 3. SADRECHA SODA | 10. DEVGHAR (PRAYER RM) | (BATH/TOILET) |
| 4. KACHERI (OFFICE) | 11. TIJORI (TREASURY) | 16. SWAYAMPAK GHAR (KITCH) |
| 5. KHALBATKHANA | 12. POTHICHI KHOLI | 17. KOTHAR (STORE) |
| (NEGOTIATION RM) | (MANUSCRIPT RM) | 18. KARANJA (FOUNTAIN) |
| 6. BAITHAK (RECEPTION) | 13. GO'THA (COW SHED) | 19. TULSI VRINDAVAN (SHRINE) |
| 7. DALAN (LIVING RM) | 14. BALANTINICHI KHOLI | 20. REAR ENTRY |

FIG. 9.

TYPICAL WADA - 2

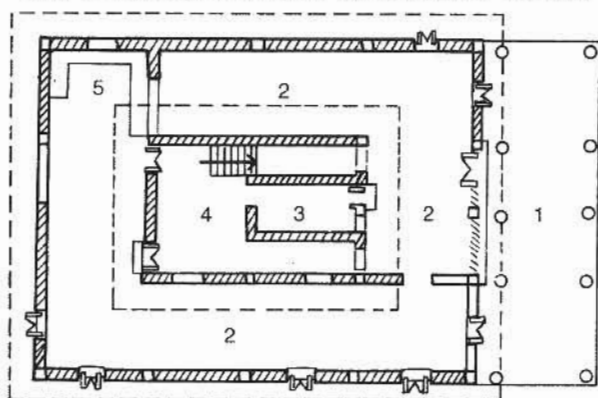


- | | | |
|-------------------|-----------------------|-----------------------|
| 1. DOOR | 10. VERANDAH | 19. HOUSEHOLD SHRINE |
| 2. ANTECHAMBER | 11. TUNNEL, TREASURY | 20. KITCHEN |
| 3. RECTANGULAR RM | 12. OFFICE STORE | 21. VERANDAH, DINING |
| 4. OFFICE | 13. STORE | 22. WATER PIPE |
| 5. FOUNTAIN | 14. STRONG RM | 23. VENTILATION |
| 6. OPEN COURTYARD | 15. MANUSCRIPT STR | 24. COW SHED |
| 7. RECTANGULAR RM | 16. MEDICINE RM | 25. SERVANT'S QUARTER |
| 8. CONFERENCE RM | 17. CHILD DELIVERY RM | 26. REAR ENTRY |
| 9. ROOM | 18. TULSI VRINDAVAN | |

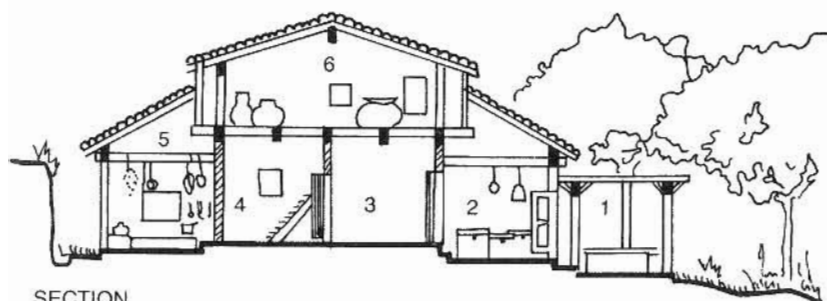
FIG. 10.

FIG. 9-10 SOURCE: KANHERE G.K 'MARATHA WADAS-A WITH OF LIFE'
MARS, 1982.

A TYPICAL HOUSE IN THE DAPOLI REGION OF KONKAN



PLAN



SECTION

- | | | | | | |
|----|-------|----|---------|----|--------------|
| 1. | ANGAN | 3. | OATI | 5. | KITCHEN |
| 2. | PADVI | 4. | MAJGHAR | 6. | MALA (ATTIC) |

FIG. 11.

SOURCE : AUTHOR'S STUDY AT DAPOLI, KONKAN.

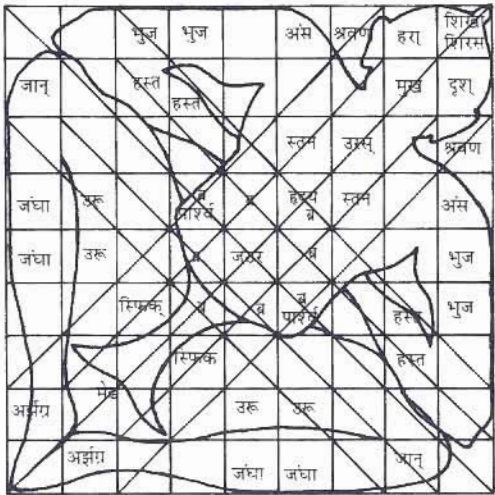


FIG. 12.

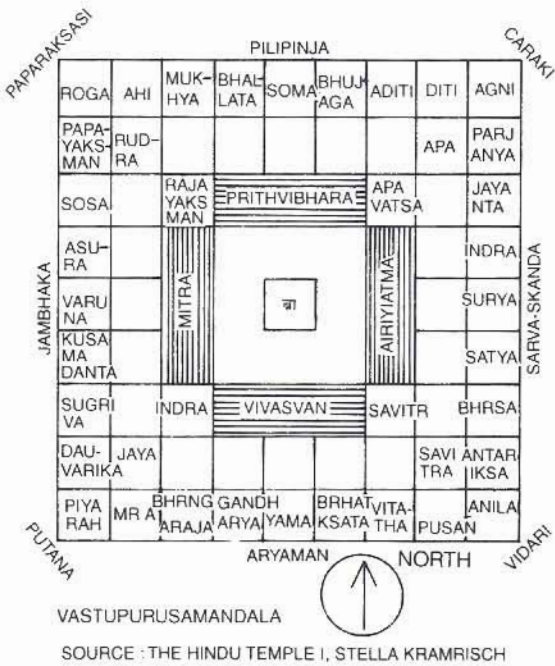


FIG. 13.

material, where and how the various functions are to be located etc., are given elaborately in the Puranas as well as in the *Vastushastra*. The Puranas also inform us that the four cardinal directions are of great importance to the planning of the house and the functional distribution of spaces must be in strict adherence to the rules stated therein to ensure the well-being of the residents. The Puranas in their available form today are conservatively said to have been written many centuries ago, but even then they show that the terms describing the typology of the houses were in use in Indian society for at least over a thousand years before the rise of the Peshwas and the so called *wada* architecture in and around Pune.

This should also mean that the typology of the houses described in the Puranas (given below) has been considered to be a guideline, a set of rules and directives for building houses for more than a thousand years. Indeed, it is difficult to imagine that these rules would be blindly followed if they were found unsuitable from a climatic or social point of view. In fact, the rules were flexible enough to be interpreted and adapted in the different climatic and geographic regions of India.

A house with a single wing was known as *ekashal griha* while a house with two wings was termed *dusopi* or *dwishal* (Figure 1). A house with rooms on the west and north was known as *yamasurya* (Figure 3) while one with rooms on east and south was called *dhanakhya* (Figure 2). A house with an internal courtyard having four wings facing each other was termed as *chatushala* or *chausopi* (Figure 7). If such a house had no door on the north, it was called *ruchaka*. A circular palace or a temple with four wings but without a west door was called *nandyavarta* (Figure 8). So the courtyard was thought to be so essential from the climatical, social and security points of view that houses were probably always conceived with inner courts. The houses with inner courtyards enclosed from three sides by verandahs and rooms were called *trishal*. While such houses without northern wings were termed *dhanyak* (Figure 6), a house without a wing on the east was termed *sukshetra* (Figure 4). The one without a wing on the west was called *pakshaghna* (Figure 5). Inevitably there were a number of myths that came to be associated with the planning principles. The entire house was conceived in terms of a square further divisible into smaller squares. Every square denoted the house of a deity (Figures 12-13). The lord of the central square is Brahma and the presiding deity of the whole site is called Vastupurusha. The north is attributed to the lord of wealth—Kubera (in the *Vastu* Kubera comes under Soma's name as Lokapala of the North); the south to the lord of death—Yama; the east to the lord of light—the Sun—Surya; and the west

to the lord of the wind—Varuna. The functional distribution of spaces was suggested in accordance with the directions. For instance, the kitchen where the fire was located, or which, in other words, was considered the source of energy, had to be in the *agneya*, the south east, which was the place of Agni³ and so on. If the functional distribution differed from these principles, naturally, this brought the wrath of a particular deity, upsetting the mental or physical well-being of one or all family members. Not only was it considered detrimental to one's prosperity and well-being—but at times even fatal—if one ignored these principles. This gave rise to a lot of myths associated with the norms laid down in the *Vastushastra*. For instance, it is said in the *Matsyapurana* that if one has a large tree right in front of the entrance of the house then it causes hatred amongst the family members, or, if a garbage dump or a place which is used by people to defecate comes right opposite the entrance of the house, then the lady of such a house commits adultery with more than five men!

The study of, or any discussion on, the myths associated with planning of buildings in accordance with the *Vastushastra*, would go beyond the scope of this paper and deserves analysis in an altogether different context. Suffice it to say at this juncture that, while most of the directives laid down in the Shastras and Puranas have a tremendous significance from the climatic and social points of view, the threats laid down in the Puranas, in case one ignored their rules, amount to psychological and social blackmail. From the climatic point of view, for instance, these rules respect the wind direction, and the sun movement in different seasons and suggest a layout of rooms so that foul smells are prevented from entering in, ensuring good air movement and sunlight within the house. From the point of view of society, the rules give the sizes of houses for different communities based on their occupation, trade and social status. The basic module of the room, verandah and courtyard acknowledges the functional and cultural requirements as have been described earlier. If one does not obey the code, he is threatened with dire consequences and misfortunes such as those that have been described above. The threat was levelled against all those who disobeyed the code, whether rich or poor or whether of high or low caste. But ironically, the codes were formulated by the upper caste men who alone had the right to study and interpret the scriptures.

Of further interest is the fact that most of the terms described above giving the typology of the houses were very much in use in the Peshwa period. A *dughai wada* meant a *wada* with a double module width

(probably with two inner courtyards having apartments on two sides). This typology offered a vast choice of arranging the rooms, verandahs, and open spaces in an ordered way. Hence, the houses could be differently planned based on the location and topography, and yet together they would exhibit an orderly pattern for a township. It was an extremely geometry-conscious measure that offered a simplistic arrangement of plots and roads. The grid pattern, which has been found suitable even for modern townships as well, and has been found in the cities of Mohenjodaro and Harappa, was easily achievable by employing this typology of houses at the city level for the purpose of city planning.

The Konkan House

While the Desh house can be called primarily an introvert type of house, the house in the Konkan is conceived in a different way. Figure 11 gives an example of a house in the Dapoli region. The Konkan house is rarely planned around a courtyard. Instead, it has the *majghar* at its centre. The *majghar* or inner room was a multipurpose room meant for private family activities. The *majghar* was usually dark and cool, surrounded by a layer of rooms around it. It was an inner core which required maximum security and insulation from the heat. Above the *majghar* was an attic where grain was stored. The Konkan region is on the western coast of India, south of Bombay. The climate is hot and humid and the monsoon heavy. This region has a rustic and green landscape rich in horticulture. Cashew, *ratamba* (*Garsenia Indica*), jack fruit (*Artocarpus Indica*), mango, coconut, *chickoo* (*Achras Zapota*), and *jамoon* (*Eugenia Jambolana*) are some of the most sought after fruit trees which are grown here. Rice, fish, and fruit are the staple diet of the Konkani. The terrain is hilly, with sweet and salt water rivulets running through the saddles and valleys to meet the Arabian sea. Most families have a patch of land where these fruit trees are grown. Hence, what is central to the day-to-day activity of the Konkani household is nurturing these trees and harvesting their yield. This is an activity which requires a lot of space, skilled people and plenty of storage. The form of the house reflects this in the most informal manner. The surrounding ground is usually terraced to hold the outdoor activities. Right from harvesting the fruits to preparing varied dishes with them, one needs to conduct activities such as drying, cutting, peeling, making by-products or utilizing the various parts of the tree such as the wood, leaves, branches etc. This activity is

manually carried out, and seems never-ending. The entire household, including the servants, have to be involved in these farm-related chores. Naturally, the space surrounding the house comes to be utilized for this purpose. Different teams of people work in all four directions, where terraces and pits (*khalas*) are created to demarcate one area from another. From the house one has an overview of all these terraces. In case of larger properties, including most farms, cattle sheds and independent storage rooms are built. Due to the heavy rains, the entire routine changes in the monsoon and people stay home. They fold up their outdoor activities and the temporary pavilions or *mandaps* and store these away. Their food in the monsoon consists of dried fish, lentils, rice, and other dried items like pickles and dry fruits. With the monsoon over, the house is only a base to be visited for meals, since all other activities, including sleeping are performed outside.

Marine life, fruit, and other horticultural produce have their own seasonal impact on the Konkani consciousness. A particular season demands the activities that precede it. This has a direct impact on the festivals as well as the routine activities at home. These all point towards a house design which cares as much for the outdoor spaces as for the indoor spaces. This rule applies even to groups of houses, where the open spaces or *angans* between and behind the houses assume importance and must be taken into consideration while locating and planning the individual houses. The concentric plan of the Konkani house seems to cater admirably to the constraints mentioned above.

The architectural design of the house is a representation of an extroverted attitude of the house. Surrounded by open space, the house does not need a courtyard open to the sky within, but rather a well protected, dry, and cool room. This is where one returns. This is what gives the feeling of being at home. The geometry of the house plan is based on convenient modules, either square or rectangular. Some of the typical houses measure 9x13, 11x15 or 13x17 arm lengths (an arm length being approximately 45 cms measured from the elbow to the finger tips).

The surrounding, open demarcated pits or *khalas*, precede the *angan*—the open yard immediately next to the house. The *angan* or the *khala* closest to the *padvi* at the entrance, is sheltered with dried coconut leaves on the top and with interwoven coconut leaves on the sides. This creates a rectangular buffer zone at the entrance which, while preventing the penetration of heat, allows filtered cool air to pass through. This is a secure place close to the house where one can work and relax. Even in the hottest April noon one can stretch out a straw mat laid on cowdung

floor and peacefully have a nap. This *angan* or *khala* covered by a *mandap* is therefore a pavilion that can be dismantled. It is a simple structure of vertical tree trunk-posts forked at the top to receive horizontal beams, also made of tree trunks. The posts are embedded in the ground, which is plastered with cowdung. The entire structure, including the roof, can be erected within hours. Before the monsoon breaks, the structure has to be dismantled and put away in a dry place for the next year. In some cases, dried coconut leaves are interwoven to create a vertical screen all around the house. This screen is an extremely economical and effective passive-cooling device in the summer. The screen cuts down the solar radiation on the surface of the walls and this effect is further enhanced by the fact that there are verandahs practically all around the house. The construction of the Konkani house is similar to that of the Desh house. It too has a timber frame and rafters, battens, and a clay tile roof of either country or Mangalore tile. The house is basically a ground-floor house with an attic or *mala* at the apex, the centre of the house. In the case of larger houses, the attic is extended to form a bedroom with projecting bracketed balconies. The walls are of either mud or laterite stone, available locally. The construction materials are available within a walkable range of the house. The construction of the houses is done by carpenters, potters, and masons, all of whom are basically farmers. In the Konkani region, as one goes closer to Goa, the Dutch and Portuguese influence is evident, especially in larger structures of temples or public buildings like markets and government offices. However, the domestic architecture remains strictly traditional.

A study of the geometry of the plans of both the Desh and the Konkani houses shows a preference for squares and rectangles. This owes as much to the construction technique and availability of building material in the region as to the rules and symbolism in the *Vastushastra* and *Vastupurushmandala*. 'The square is literally the fundamental form of Indian Architecture... It presupposes the circle and results from it. Expanding energy shapes the circle from the centre; it is established in the shape of the square. The circle and curve belong to life in its growth and movement. The square is the mark of order, of finality to the expanding life, its forms; and of perfection beyond life and death'... 'The square symbol of the extended world in its order has precedence over the circle of time... ..the square, is the larger, the comprehensive form, for it contains the cycles of measurable time.'⁴

The myth of the Vastupurusha (Figure 12) has further penetrated deep into the Indian psyche. 'The 'Br̥hat Samhitā' (LII. 2-3) narrates:

Once there was some existing thing (*satva*) not defined by name, unknown in its proper form, it blocked heaven and earth; seeing that, the Devas (gods) seized it of a sudden and laid it on the earth face downwards. In the same position as they were when they seized it, the Devas stayed on it where it lay. Brahmā made it full of gods and called it Vāstupuruṣa...⁵⁵ Cosmogonically, this is a story of the first appearance of Existence, and there are many versions available of it in the Puranas. Strictly followed in the case of temples, it also becomes a principle in planning palaces and common men's houses alike. In all types of *Vastupurushamandalas* (Figure 13), the modular division of the square is implied. The centre is given great importance. It is considered to be the place of Lord Brahma. He occupies the position from the heart to the belly of the Vastupuruṣa. 'The place of realisation of the Supreme Brahman, the centre of the Vāstupuruṣa, is assigned to Brahmā who is the effected... Brahman thus is the subtle state of manifestation... The place of Brahman (Brahmāsthāna) corresponding to Brahmapura in the universe is the Hiranyagarbha, the Embryo of Splendor, the primordial germ of cosmic light. Similarly, from the Brahmāsthāna proceeds the light of all times and in every direction;... and encompasses the extent of corporeal manifestation.'⁵⁶ This shows how important the centre of the house is to the Indian psyche. It is the *kunda* where life originates, and it is the place of the lord of creation—Brahma. That is also why the Konkani house has the *majghar* at its centre, while the Desh house has a courtyard at its centre. Both the plan types are developed from the square. While one has an open square at its centre, the other has the most important inner room. Both, therefore, are expressions of the same Vastupuruṣa principle. Architecturally and spatially, while the house in the Desh expresses the introvert aspect of the Marathi house, the Konkani house is most eloquent about its extrovert nature. These two, as it were, are two sides of the same coin, in that they carry forward the myth of the Vastupuruṣa and the modular geometry of the square, as the symbolic meaning of the manifested form of cosmic order.

In the times that we live in today, the evolutionary trends in construction, crafts and concepts of planning are being brushed aside by the achievements in technology and science. The principles of planning which were based not only on myths but also on a sound knowledge of centuries together of the climatology, ecology and environment, which few have the patience to study, are being forgotten, at least temporarily. However, these principles of planning, which do not necessarily prompt a rigid architectural expression, will always be reverted to, because they

are capable of generating contemporary architecture most appropriately suited to the Indian psyche, environment and climate.

Notes

1. Dingle, Narendra, 'Culture & Space'. *Architecture + Design*, vol X, No 6 (Nov-Dec 1993), New Delhi.
2. See V.L. Manjul's essay 'House and Surroundings in Jñanesvari', in this volume.
3. 'Agni, amongst the gods, is assigned the SE; Agni, amongst the presiding divinities of the stars, is assigned the NE. The latter position he holds in the Vāstumaṇḍala of the 'Bṛhat Saṃhitā' as Padadevatā.' (Stella Kramrisch, *The Hindu Temple I*, (Delhi: Motilal Banarsidass, 1976), 33. 'But for this and the corresponding changes in the intermediate directions, where Agni occupies the South-East in the place of Anila... who is Māruta who is Vāyu, the wind, is placed in the North-Western corner, there are no major differences in the positions of the Devatās in the Vāstusāstra of all the schools and at all times.' (footnote 30 on the same page).
4. Kramrisch, *The Hindu Temple I*, 22 and 41.
5. *ibid*: 73.
6. *ibid*: 88-9.